

Jayesh Vadge

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WORK EXPERIENCE

AI/ML Engineer, Accenture APR 2023 - present

- Architect, scale, and maintain stateful multi-agent frameworks, advanced RAG conversation engines, and local LLM microservices to automate complex operational workflows for global Telecom clients.
- Translate corporate product specifications into technical documentation, system architectures, and secure backend API boundaries using Python and FastAPI.
- Built Retrieval-Augmented Generation (RAG) workflows using embedding models, semantic search techniques, and vector-based retrieval mechanisms for knowledge-driven applications.
- Worked with Generative AI technologies including LLMs, Prompt Engineering, LangChain, and agent-based workflows to automate business processes and enhance user experiences.
- Build autonomous software agents using dynamic function calling to securely bridge natural language processing models with relational databases and core network APIs.
- Collaborated with business stakeholders, engineers, and cross-functional teams to translate business requirements into scalable AI solutions deployed on cloud platforms and LLMOps environments.

Quality Assurance Engineer, Accenture OCT 2022 - MAR 2023

- Planned and executed test cases to ensure software quality, system reliability, and requirement validation for enterprise applications.
- Collaborated with developers and business teams to identify defects, improve workflows, and support Agile project delivery.
- Worked with SQL queries, API testing, and automation workflows to improve software stability and operational efficiency.

SKILLS

Generative AI:	LLMs, Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Function Calling, Tool Use, System Prompt Engineering, Few-Shot Learning, Reasoning and Acting (ReAct)
Frameworks:	LangGraph, CrewAI, LangChain, LlamaIndex, NVIDIA NeMo Guardrails, PyTorch, Scikit-learn, Jupyter Notebook
Machine Learning:	Machine Learning, Deep Learning, NLP, Predictive Modeling, Classification, Regression, Clustering, Feature Engineering, XGBoost, LightGBM, CatBoost, BERT
Models and Vector DBs:	GPT-4o, Claude 3.5 Sonnet, Llama 3, Ollama, Pinecone, ChromaDB, Hugging Face
Software & MLOps:	Python, SQL, FastAPI, Streamlit, Pandas, Git, Docker, Azure

EDUCATION

2018 - 2022	Bsc (Computer Science) at Savitribai Phule Pune University	(CGPA: 7.85/10.00)
2024 - 2026	MBA (Business analytics) at Savitribai Phule Pune University	(CGPA: 7.81/10.00)

PROJECTS

- **Multi-Agent Telecom Support Assistant :**

Problem: Telecom support operations faced severe bottlenecks due to manual ticket classification, slow root cause identification, and inconsistent response accuracy.

Solution: Developed an Agentic AI system using LangGraph to coordinate specialized agents for classification, root cause analysis, and response generation. Integrated an advanced RAG architecture using vector embeddings for semantic knowledge retrieval.

Impact: Reduced manual support overhead significantly while substantially increasing customer response accuracy and system execution velocity.

- **Next-Gen Retrieval-Augmented Generation (RAG) Platform :**

Problem: Support staff faced elevated call durations due to manual lookups across thousands of pages of unstructured internal roaming and device compliance PDFs.

Solution: Built an enterprise RAG platform using LangChain and Pinecone. Integrated an advanced Cross-Encoder Reranker structure alongside custom spaCy tokenization layers optimized for specialized telecommunications acronyms.

Impact: Boosted response accuracy by 35% and drastically accelerated onboarding training velocity for incoming customer support cohorts.

- **Agentic Billing Dispute and Compensation Workspace :**

Problem: Billing adjustments require absolute mathematical precision, a standard where baseline LLMs frequently hallucinate during long-context generation.

Solution: Designed a ReAct agent architecture via LangChain utilizing few-shot prompts. Equipped the model with a secure Python Interpreter tool to isolate numerical computation, adding NVIDIA NeMo Guardrails for PII data masking.

Impact: Processed over 500,000 monthly queries autonomously, achieving an 18% increase in First Contact Resolution (FCR) metrics.

- **Localized GenAI Test Case Generation Engine :**

Problem: QA teams spent valuable hours manually parsing Requirement Documents, but strict compliance guidelines blocked uploading data to public cloud APIs.

Solution: Developed an air-gapped web application powered by Ollama, hosting Llama 3 (8B) locally on corporate GPU clusters. Integrated ChromaDB vector stores and a Streamlit user interface to automatically generate fully structured test suites.

Impact: Accelerated release velocities by reducing manual test case design lifecycles by 60% with total data privacy.

ACHIEVEMENTS AND AWARDS

- **Apex Award FY23 Q2/Q3** - Extra-Miler is given to individual who has gone over and beyond client deliverables and stepped up to support the team/peers in times of need)
- **Spotlight Award** - Awarded with the 'Spotlight Award' for two consecutive years(Jan'24,March'25) at account level for Excellent work in projects.

CERTIFICATIONS

- **Microsoft Certified: Machine Learning Operations Engineer Associate**